

Designed by Remus Teodorescu 2026

LECTURE 11.2

Obstacle Avoidance

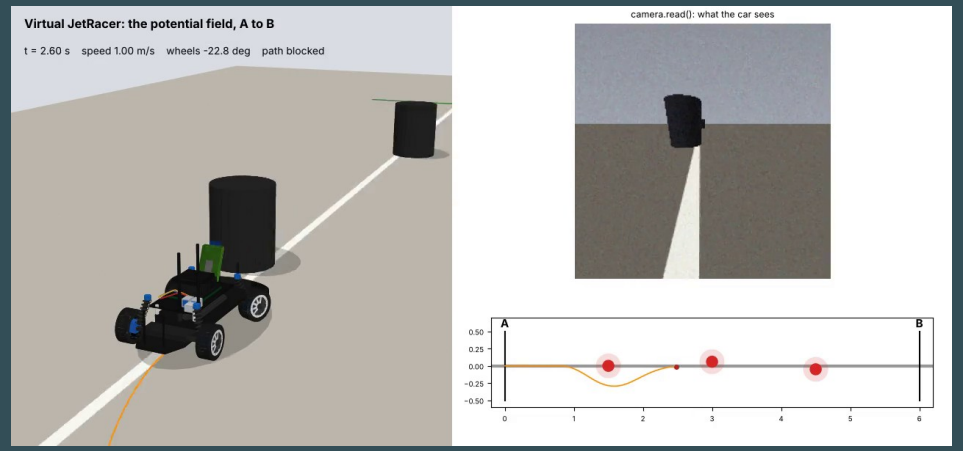
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

IDENTIFY · DECIDE · IN TIME



Something stands in the path. The lecture answers two questions, first in words and then in mathematics: how do I identify the obstacle - a network that tells free from blocked, and what its errors cost - and how do I decide the action - stop, steer around or follow a field - in time?

Learning Objectives

SECTIONS, IN ORDER

1. **Identifying obstacles** — what blocks the path, and how a network tells
2. **Deciding the action** — stop, steer around, or follow a field
3. **Reacting in time** — the stopping distance, and a faster loop

Each section answers its question twice: first in words, then in mathematics. The example is Ex_11's drawn frames and its A-to-B course, where the truth is known; the real JetRacer drives the same task in a lab demonstration.

READ, AND WHAT EACH ANSWERS

UDL ch. 5	how do I identify an obstacle?
UDL ch. 19	how do I decide the action?
PyTorch	how fast must the loop be?

Identifying an Obstacle

Obstacle — anything that stands in the path ahead, near enough that the car must act on it. Its colour and shape do not decide it; where it stands does. A shadow on the floor is dark but solid nothing, so it is not an obstacle.

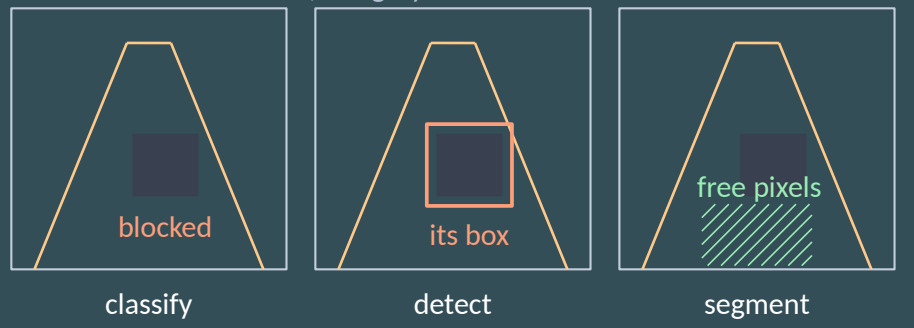
THREE QUESTIONS ABOUT EVERY FRAME

is it in the lane?	beside the path is not in the way
is it near?	far away can wait for the next frame
is it solid?	a shadow or a reflection is not

TALK · 2 MIN How would you decide, in your own words, what counts as an obstacle for a car? Then, in the three ways on the right, which one tells the car where to steer?

THREE WAYS TO SEE IT

the lane ahead in amber; the grey block an obstacle on it



	the output	enough to
classify	free or blocked	stop or go
detect	a box: where, how big	steer around
segment	every free pixel	plan a path

Ex_11 classifies, as NVIDIA's JetBot does, turning away while the frame reads blocked.

Image Classification

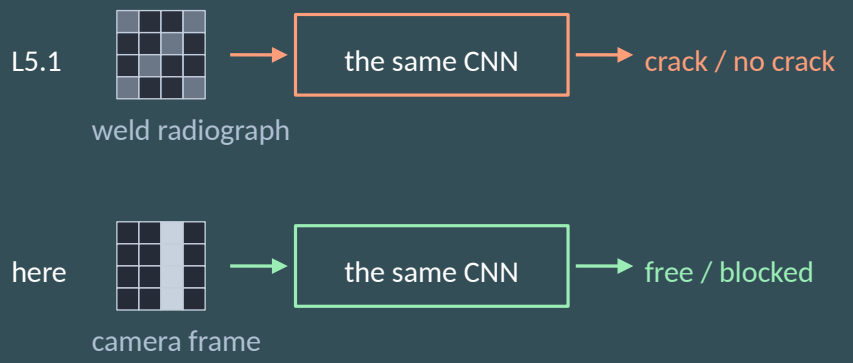
Image classification — giving a whole image one label from a fixed set. Part 1 built every piece this lecture needs: the convolutional network that reads the image, the softmax that turns its outputs into a probability, and the cross-entropy loss that trains them.

EVERY PIECE COMES FROM PART 1

the features	convolution kernels learned from frames - L5.1
the probability	the softmax over the two outputs - L4.2
the loss	cross-entropy on labelled frames - L6.1
the reuse	a trained backbone kept - L6.2

New in this lecture is only what the frames show, and what an error costs.

THE SAME NETWORK, A NEW QUESTION



L5.1's network classified weld radiographs, crack or no crack. The same architecture here answers free or blocked: 224 × 224 pixels in, two numbers out. What changed is the question, the frames, and what an error costs - the pattern of this whole half of the course: Part 1's methods, pointed at an engineering problem.

Free or Blocked

Binary classification — a network with two outputs, one for free and one for blocked, turned into a probability by the softmax. It is trained with cross-entropy on frames labelled by where the obstacle stands, so the network learns what makes a frame blocked.

THE PROBABILITY, AND THE LOSS THAT TRAINS IT

$$p_{\mathrm{b}} = \frac{e^{z_{\mathrm{b}}}}{e^{z_{\mathrm{f}}} + e^{z_{\mathrm{b}}}}$$

$z_{\mathrm{f}}, z_{\mathrm{b}}$ the network's two outputs, p_{b} the probability of blocked

$$\mathcal{L} = -\frac{1}{N}\sum_{k=1}^{N}\left[y_k \ln p_{\mathrm{b},k} + (1-y_k)\ln(1-p_{\mathrm{b},k})\right]$$

y_k 1 when frame k is blocked, 0 when free, N the frames in a batch

THE HEAD AND THE LOSS

```
head = torch.nn.Linear(32, 2)                    # two outputs: free, blocked
p_b = torch.softmax(net(x), dim=1)[:, 1]         # the probability of blocked
loss = torch.nn.functional.cross_entropy(net(x), y)   # the loss above
```

ON 800 TEST FRAMES

	dark rule	network
right	85.1 %	96.6 %
obstacles missed	19.7 %	14.6 %
false stops	13.5 %	0.2 %
grey boxes missed	70.5 %	43.2 %

The dark rule calls anything darker than the floor in the lane blocked, so shadows stop the car. The network, trained on frames with both, does not stop for them; a grey box, as light as the floor, slips past both.

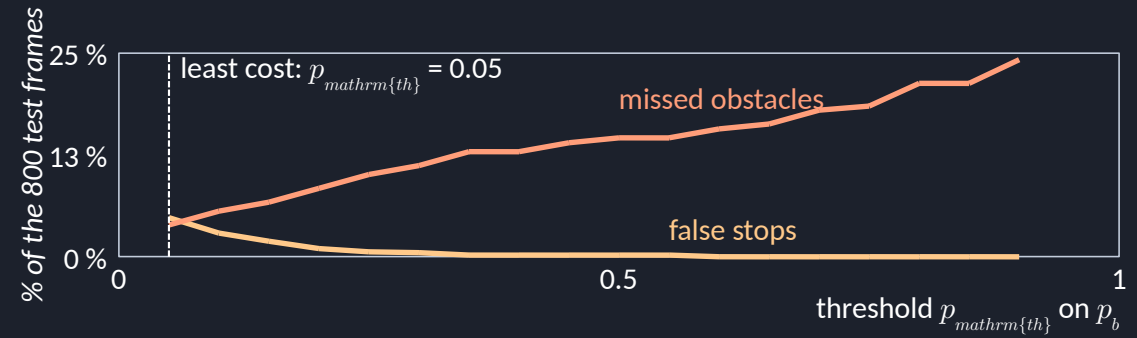
The Cost of an Error

Decision threshold — the probability $p_{mathrm{th}}$ above which the car treats the path as blocked. A missed obstacle costs a collision; a false stop costs a second. The two errors are not equal, so $p_{mathrm{th}}$ is set by their costs, not at one half.

THE EXPECTED COST OF A THRESHOLD

$$C(p_{\text{th}}) = c_{\text{miss}}\, P_{\text{miss}}(p_{\text{th}}) + c_{\text{stop}}\, P_{\text{stop}}(p_{\text{th}})$$

p_{th} the threshold on p_{b}, $P_{\text{miss}}, P_{\text{stop}}$ the shares of missed obstacles and false stops
$c_{\text{miss}} \gg c_{\text{stop}}$: what each costs, a collision against a lost second



CHOOSING THE THRESHOLD

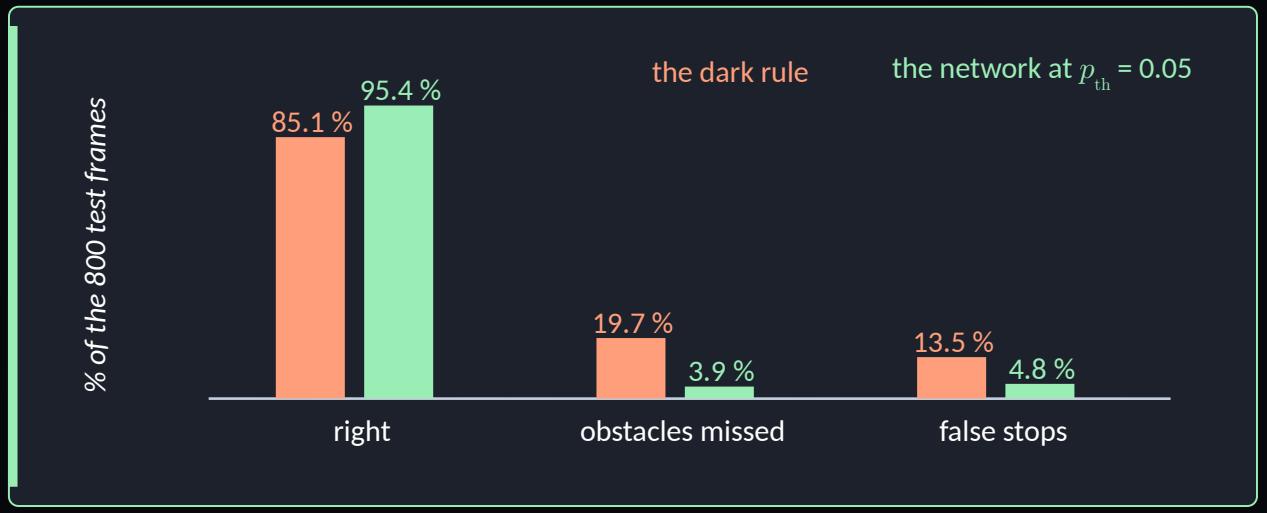
a miss	a collision: the run is lost
a false stop	a pause: a second is lost
so	set $p_{mathrm{th}}$ low, accept false stops
the twin	14.6 % missed at 0.5, 3.9 % at 0.05
on the car	measure both again

Lowering $p_{mathrm{th}}$ trades missed obstacles for false stops; the costs say where to stop.

What the Network Buys

Learned robustness — what a classifier gains by being trained on the disturbances it will meet. The dark rule's errors are fixed in its steps, chosen once by an engineer; the network's fall with every labelled frame added. The gain is measured on the same test frames, never assumed.

RULE AGAINST NETWORK, AT THE CHOSEN THRESHOLD



WHERE THE GAIN COMES FROM

the rule	one fixed step: dark means obstacle
so a shadow	dark, and stops the rule's car
the network	trained on shadows and glare
more frames	its errors fall; the rule's cannot

CONCLUSION

The network wins where the rule's assumption breaks: a shadow is dark but not an obstacle. What neither sees, the grey box as light as the floor, is a limit of one camera; the mini projects add a second sensor for it.

Transfer Learning

Transfer learning — reusing a network trained on one task as the start of another. A ResNet-18 trained on ImageNet already detects edges, textures and shapes, so on the car only its head is new, and a few hundred frames of free and blocked are enough.

THE ON-CAR RECIPE, IN FOUR LINES

BACKBONE KEPT, HEAD REPLACED

```
net = torchvision.models.resnet18(pretrained=True)   # the backbone, kept
net.fc = torch.nn.Linear(512, 2)                     # the new head, two outputs
x = normalize(frame, IMAGENET_MEAN, IMAGENET_STD)    # the backbone's statistics
opt = torch.optim.Adam(net.parameters(), lr=1e-4)    # fine-tune it all
```

Ex_11's small network trains from nothing, because its frames are drawn in thousands; on the car a few hundred real frames meet a reused backbone.

THE SILENT ERROR

forget it	and the model still trains
the loss	still falls
the accuracy	mediocre, for no visible reason
so	it is the first thing to check

The backbone expects its own mean and standard deviation, channel by channel; fed raw pixels it sees every frame through the wrong glass.

Deciding the Action

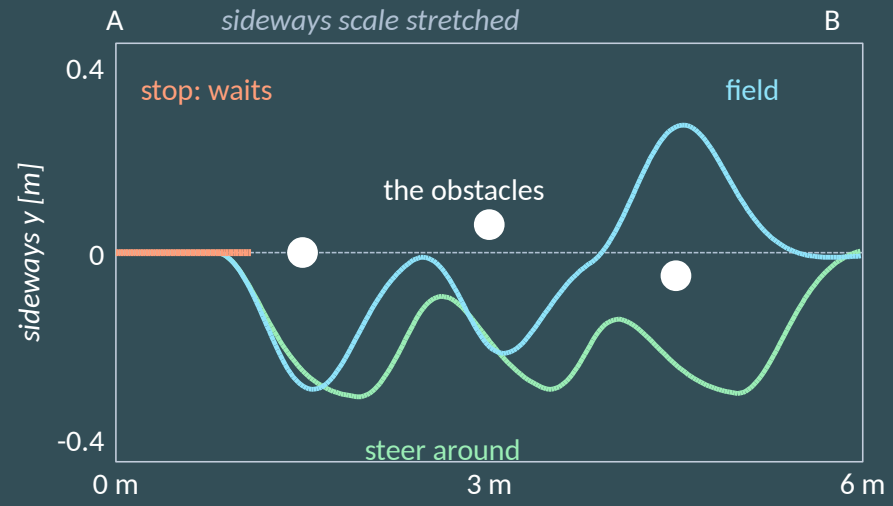
Avoidance strategy — the rule that turns what the car sees into what it does: stop, steer around, or follow a field around the obstacle. It is chosen first, because it decides what the network must tell: that, which side, or where.

THE STRATEGIES, AND WHAT EACH NEEDS

strategy	the car needs	reaches B?
stop	free or blocked	not past an obstacle that stays
steer around	blocked, and which side	yes, slowed down
field	where the obstacle is	yes, with a smooth path

TALK · 2 MIN How would you decide, in your own words, between stopping and steering around? Then, in the paths on the right, why does stopping never reach B?

THE THREE ON EX_11'S COURSE, 10 HZ



Three obstacles on a 6 m course, each 16 cm across; the car is 20 cm wide. Course and sizes assumed.

Steering Around

Avoidance rule — the decision written as mathematics. The steer-around rule switches the lane aimed at when the network says blocked; the potential field adds a push away from every obstacle to the pull back to the line, and the car aims where the sum is lowest.

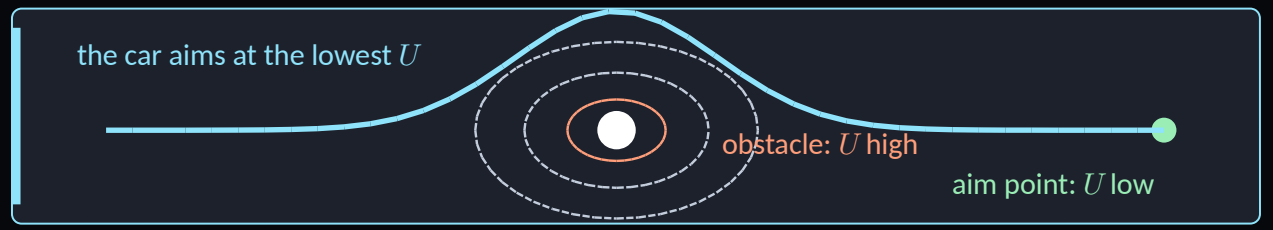
THE RULE, AND THE FIELD

$$y_{\text{ref}} = \pm(0.30\ \text{m})\, s, \qquad v = v_0\,(1 - 0.4\, s)$$

s the verdict (a flag, not a time): 1 when $p_b \geq p_{th}$, 0 when free, v_0 the cruising speed
y_{ref} the side offset of the lane aimed at, away from the obstacle

$$U(y) = y^2 + \sum_i \frac{K}{d_i(y)}, \qquad y^* = \arg\min_y U(y)$$

y the sideways position one aim distance ahead, $d_i(y)$ the gap to obstacle i
K the strength of the push, y^* the point the car aims at: the lowest U



MEASURED ON THE COURSE

	time to B	closest gap
stop, 10 Hz	never	22.2 cm
steer, 5 Hz	8.4 s	2.3 cm
steer, 10 Hz	8.3 s	0.6 cm
field, 10 Hz	6.3 s	6.7 cm
field, 30 Hz	6.2 s	3.7 cm

A negative gap is a collision. The field is faster because it never slows down, but it needs where the obstacle is - a detector or a range sensor - not only free or blocked.

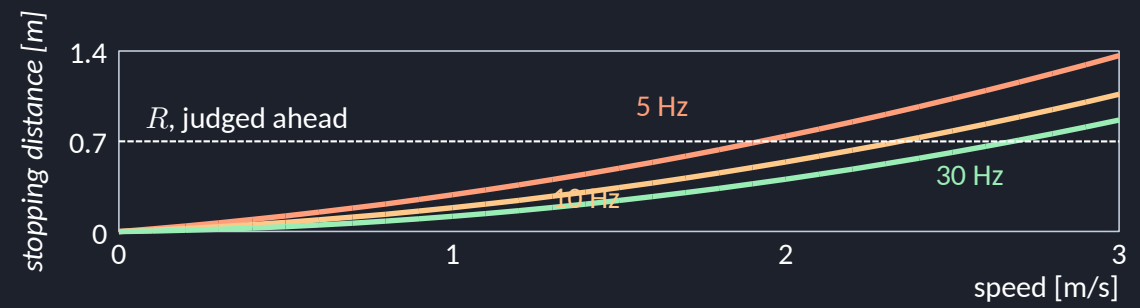
Stopping Distance

Stopping distance — the distance the car covers from the frame that shows the obstacle to standing still: the blind distance of one decision, then the braking the tyres allow. It must be shorter than the range ahead that the camera judges.

THE CONDITION FOR A SAFE SPEED

$$d_{\text{stop}} = \frac{v}{f} + \frac{v^2}{2\mu g} \leq R$$

v the speed, f decisions per second, μ the tyres' friction
R how far ahead the camera judges free or blocked



THE SAFE SPEED, BY LOOP RATE

5 Hz	1.93 m/s
10 Hz	2.34 m/s
30 Hz	2.68 m/s
range R	0.70 m, assumed
friction μ	0.6, assumed

A faster loop buys speed only up to the braking limit; a longer range buys more.

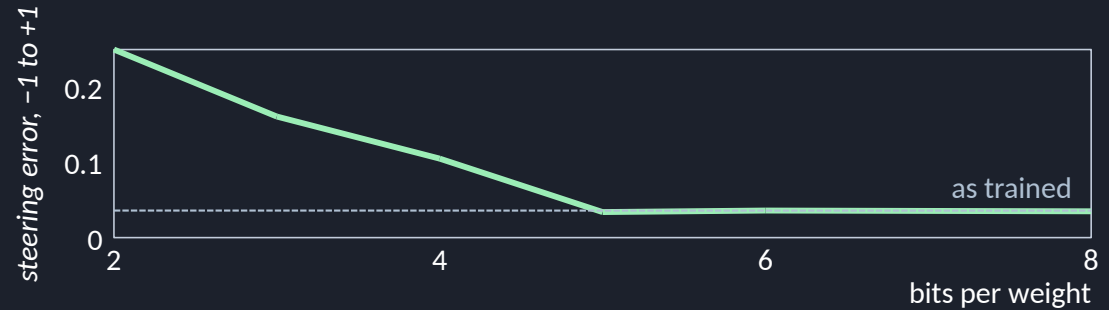
Deployment on the Car

Quantisation — storing each weight in b bits instead of 32: the range of the weights is cut into 2^b steps and every weight is rounded to its step. With compilation for the one GPU on the car, it is what makes the loop faster.

THE ROUNDING, AND WHERE IT BREAKS

$$w_b = \Delta \,\mathrm{round}\left(\frac{w}{\Delta}\right), \qquad \Delta = \frac{w_{\max} - w_{\min}}{2^b - 1}$$

w a weight, w_b the weight stored in b bits, Δ the step between the 2^b values kept



COMPILED FOR THE DEVICE

train	on a laptop or Colab
move	the weights only, not the model
convert	torch2trt, on the car itself
half precision	the large win on this GPU
measure	the whole loop, before and after

Five bits still steer; four do not. Find the cliff by measuring.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
object detection	a box and a class for each object	UDL ch. 10	crowded scenes, people
semantic segmentation	every free pixel of the floor	UDL ch. 10	planning a path
path planning	a route on a map, not a reflex	UDL ch. 19	warehouses, road cars
learned policies	the action learned from reward	UDL ch. 19	simulators, legged robots
sensor fusion	a LiDAR and an ultrasonic ranger too	UDL ch. 10	mini project B

WHY THEY ARE LEFT OUT

Ex_11 classifies free or blocked from one camera on a known course, so none of them is needed to do it. The last row is mini project B: the camera fused with a LiDAR and an ultrasonic ranger.

Exercise 11

THE SAME NOTEBOOK · HERE: SECTIONS 5 – 6, THE REPORT

- **5** — free or blocked; TODO 2, the loss and the threshold
- **6** — stop, steer around, field; TODO 3, the potential field
- **7** — fewer bits, this lecture's part of the loop section
- **8 – 11** — the race, the report and the mini projects
- **1 – 4** — L11.1's half: run them first, the networks carry over



the same task on the real car: a demonstration in the lab

THE QUESTION EACH PART ANSWERS

5	is the path blocked, at what cost?
6	which strategy reaches B first?
7	how few bits still steer?
L11.1	sections 1 – 4: where is the path?

REPORT QUESTION, TO CONCLUDE

Which avoidance strategy reaches B fastest without a collision, and what limits its speed: the errors of the networks, the loop rate, or the distance needed to react?

SUMMARY

Key Takeaways

OBSTACLE CLASSIFIER

what it is	a CNN with two outputs, free and blocked
function	the probability that the path is blocked
train, tune	cross-entropy; the threshold by the costs
vs a dark rule	not fooled by shadows, misses fewer
used for	the first input of every strategy

AVOIDANCE POLICY

what it is	the rule from what is seen to what is done
function	stop, steer around, follow a field
train, tune	chosen first: it sets what the network tells
vs stopping	reaches B; needs which side or where
used for	a safe speed: $v/f + v^2/(2\mu g) \leq R$

Next — L12.1: the power grid, where a network estimates what the meters cannot see.

Questions

1 – 2 · IDENTIFYING AN OBSTACLE

1 **How can a network decide from one camera frame whether the path is blocked, where a dark-pixel rule fails?**

The network has two outputs, free and blocked, turned into a probability by the softmax and trained with cross-entropy on frames labelled by where the obstacle stands: in the lane and near. A rule that calls anything darker than the floor blocked stops for shadows and misses grey obstacles. On the rendered frames the rule is right 85 % of the time and the network 97 %, almost never stopping for a shadow; the rule misses seven grey boxes in ten, the network four. *UDL ch. 5; Ex_11 section 5*

2 **Why is the threshold on the probability of a blocked path not set at one half, and how is it chosen?**

A missed obstacle costs a collision and a false stop costs a second, so the two errors have very different costs. Lowering the threshold trades missed obstacles for false stops, and the threshold is chosen where the expected cost, each error's rate times its cost, is smallest. On the rendered frames the network misses 15 % of the obstacles at one half and 4 % at 0.05, where it stops for 5 % of the free frames; on the car both rates are measured again. *UDL ch. 5; Ex_11 section 5*

3 – 4 · DECIDING THE ACTION

3 **How does the avoidance strategy decide what the network must tell, and why does stopping alone not reach B?**

Stopping needs only free or blocked, but an obstacle that stays keeps the car standing, so it never reaches B. Steering around needs which side is free, and shifts the lane aimed at while the path is blocked, slowed to 60 %. A potential field adds a push away from every obstacle to the pull of the aim point, so it needs where each obstacle is, from a detector or a range sensor. On the course the field reaches B in 6.3 s, steering around in 8.3 s with half a centimetre to spare. *UDL ch. 19; Ex_11 section 6*

4 **Why does the rate of the control loop limit the safe speed of a car that avoids obstacles?**

From the frame that shows the obstacle to standing still, the car drives one blind distance, its speed divided by the decisions per second, and then brakes as far as the friction of the tyres allows. That sum must stay within the range the camera judges. With a range of 0.7 m the safe speed is 1.9 m/s at 5 decisions a second and 2.7 m/s at 30; a faster network shortens only the first of the two distances. *Ex_11 section 7*

NEXT

L12.1 — Power Grid Stability Estimation

The last block: the power grid, where a network proposes the state and the power flow equations judge it - with no true state to train on.

BEFORE THE NEXT SESSION

read	UDL ch. 5
run	Ex_11's notebook, sections 5 to 7
bring	your strategy's time to B, and its gap
mini projects	faster avoidance; LiDAR and sonar

Block eleven is finished: a car that finds its path and a car that gets past what blocks it.